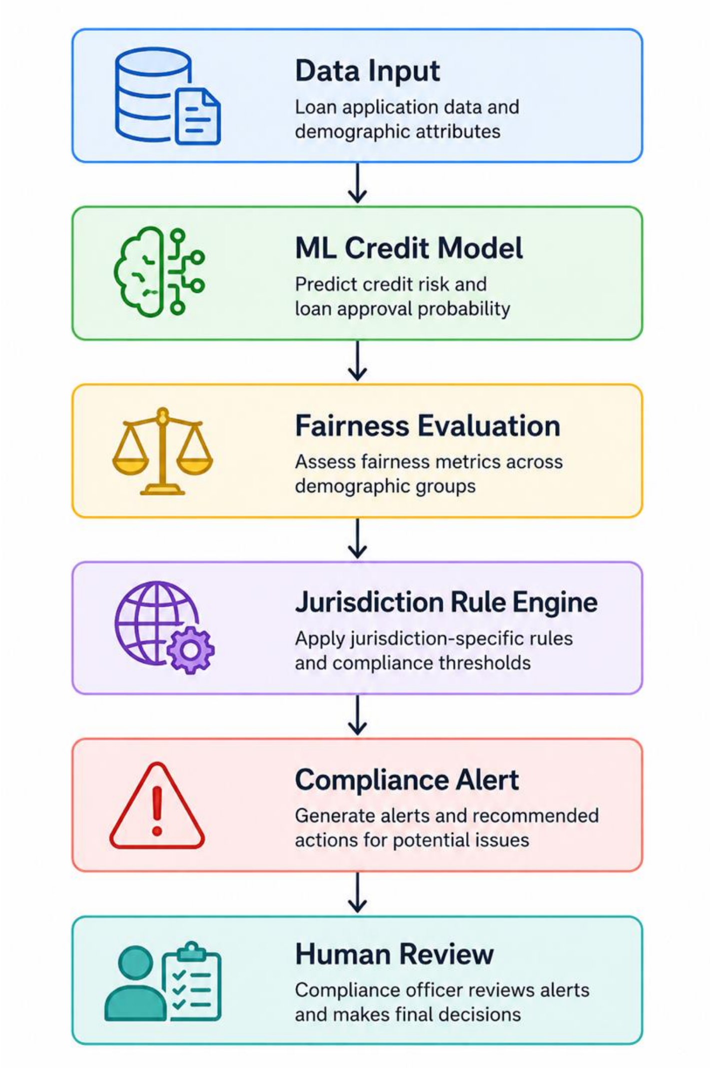
Project overview

Jurisdiction-Aware AI Fair Lending Monitoring Tool



- **Problem**
  - AI-based lending systems may unintentionally generate discriminatory outcomes.
- **Selected Entity**
  - HSBC
- **Jurisdictions**
  - United States vs European Union
- **Domain**
  - AI credit scoring and fair lending governance

Regulatory Comparison

US vs EU regulatory logic comparison

Category	United States	European Union
Main Focus	Fair lending monitoring	High-risk AI governance
Regulation	Regulation B / ECOA	EU AI Act
Key Requirement	Adverse action explanation	Bias testing + governance
Threshold Example	0.10 disparity	0.05 disparity
Compliance Outcome	Pass	Breach

The same model output may trigger different compliance outcomes under different jurisdictions.
Regulatory expectations differ substantially across jurisdictions, requiring compliance systems to support localized governance logic.

Technical Prototype

- **Dataset**
- German Credit Dataset with synthetic demographic augmentation
- **Model**
- Logistic Regression
- **Why Logistic Regression?**

1. Interpretable
2. Common in regulated financial environments
3. Easier for explainability and governance review

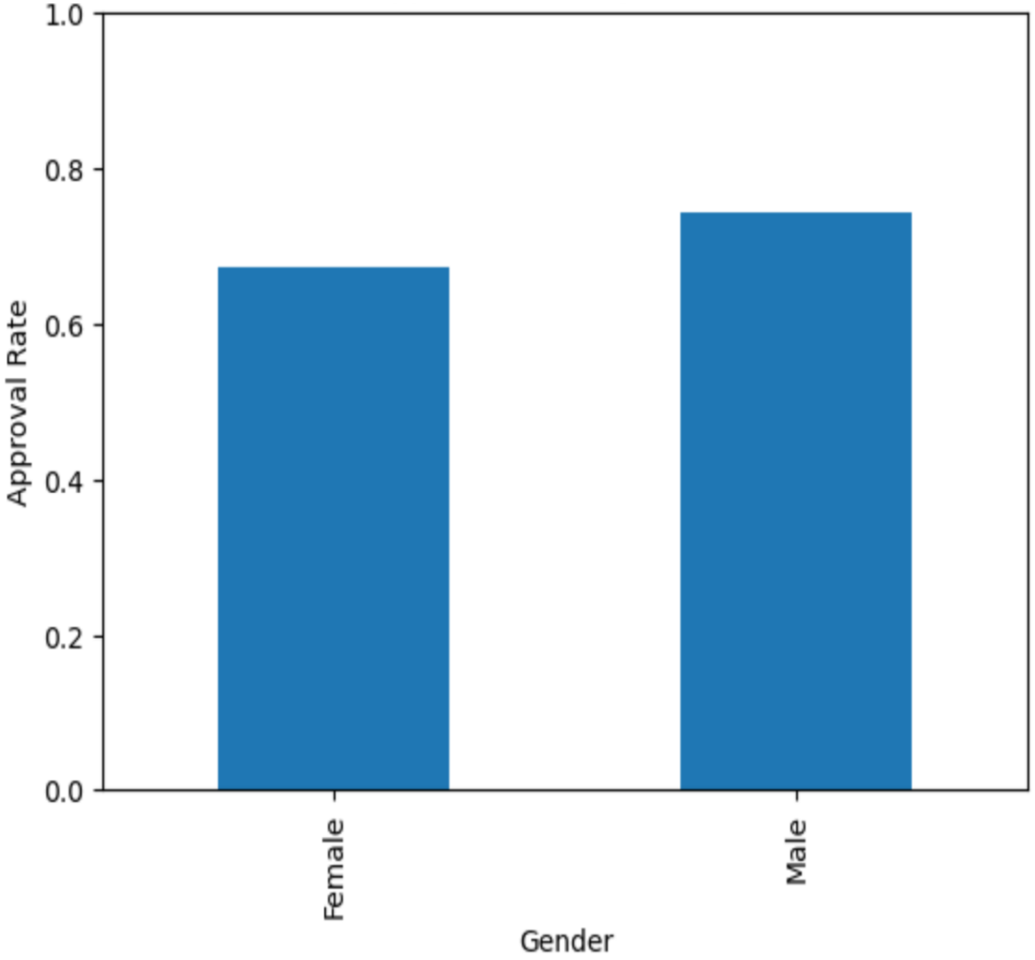
```
Accuracy: 0.7533333333333333
ROC-AUC: 0.7759788359788361

Classification Report:
              precision    recall  f1-score   support

           0       0.61      0.48      0.54        90
           1       0.80      0.87      0.83       210

    accuracy                           0.75       300
   macro avg       0.70      0.67      0.68       300
weighted avg       0.74      0.75      0.74       300
```

Prototype Scope

- Academic prototype
- Synthetic demographic augmentation
- Not intended for real-world lending deployment
- Designed for governance experimentation

Fairness Evaluation & Compliance Result

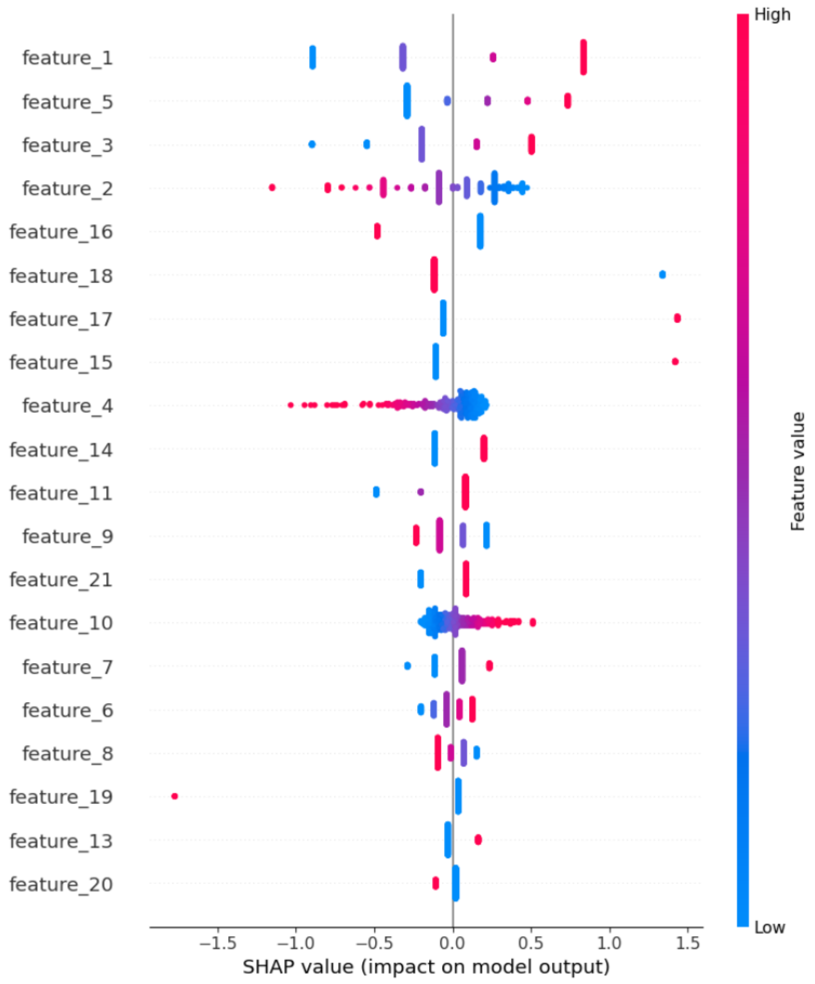
Approval Rate by Gender



jurisdiction	threshold	disparity	status
US	0.10	0.07	Pass
EU	0.05	0.07	Breach
Observed approval disparity = 0.0708			

The EU threshold is stricter than the US threshold, therefore the same disparity value generates different compliance outcomes.

Governance, Limitations & Conclusion



SHAP Explainability： Supports transparency and adverse action reasoning.
Human Review： The tool supports compliance officers rather than replacing them.

Limitations:

- Synthetic data augmentation
- Simplified regulatory thresholds
- No real-time regulatory API integration
- Prototype only, not production-ready

AI governance systems must adapt to jurisdiction-specific regulatory expectations rather than applying a single global compliance standard.

Future Improvements

Governance

- Real-time regulatory updates
- Dynamic jurisdiction configuration
- Expanded fairness metrics

Technical

- Larger datasets
- Additional ML models
- Drift monitoring
- Real-time compliance dashboard

Operational

- API integration
- Human escalation workflow
- Audit logging system

Future versions should support scalable multi-jurisdiction AI governance workflows.